MILITARY SPECIFICATION

ELECTRONIC COMPONENTS ASSEMBLY (11691622)

This specification is approved for use by the Rock Island Arsenal, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of electronic components assembly (ECA) for use in the M28A1E1 Helicopter Armament Subsystem.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-W-13855	-	Weapons: Small Arms and Aircraft Armament Subsystems, General Specification for
MIL-P-14232	-	Parts, Equipment and Tools for Army Materiel, Packaging and Packing of
MIL-I-45607	-	Inspection Equipment, Acquisition, Maintenance and Disposition of

STANDARDS

Federal

Fed. Std. No. 356	-	Commercial Packaging of Supplies and Equipment

Military

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-721	-	Definitions of Effectiveness Terms for Reliability, Maintainability, Human Factors and Safety
MIL-STD-781	-	Reliability Test Exponential Distribution

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DRAWINGS

Rock Island Arsenal		
F11691622	-	Electronic Component Assembly

(Product drawings referenced in this specification form a part of F11691622.)

PUBLICATIONS

Rock Island Arsenal		
P11691622	-	Packaging Data Sheet for Electronic Components Assembly
QE-G-DL 11691622	-	Quality Engineering Data List for Electronic Components Assembly

(Acceptance inspection equipment drawings referenced in this specification form a part of QE-G-DL11691622.)

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article. Requirements for submission of first article shall be as specified in the contract (see 6.2). Unless otherwise specified in the contract (see 6.2), the first article shall include the pilot pack (see 5.1).

3.2 Materials, construction and design. The electronic components assembly (ECA) shall conform to the materials, construction and design requirements specified herein, on Drawing F11691622 and MIL-W-13855.

3.2.1 Weight. The weight of the ECA shall not exceed 13.6 pounds.

3.3 Functional requirements.

3.3.1 Power input. The ECA shall perform as specified herein when supplied with 22 to 30 volts direct current (vdc) from the aircraft power supply or equivalent, and with 115 + 4.5 - 7.5 volts alternating current (vac), 400 ± 20 Hertz (Hz) from the aircraft inverter. Maximum current shall be 3.5 amperes. Unit shall function from 22 to 24 vdc with no more than 10 percent degradation of performance. Testing shall be as specified in 4.5.3.2.

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3.3.2 DC - DC converter. Converters (A1A1, A1A2) shall convert aircraft supplied 22 to 30 vdc into a regulated +28 ± .12 and -28 ± .12 vdc. When the input power supply drops to 22 vdc, the regulated output shall be no less than +25.2 vdc and no more than -25.2 vdc. Testing shall be as specified in 4.5.3.2.

3.3.3 Power transformer. The power transformer shall transform the aircraft converter supply of 115 + 4.5 - 7.5 vac, 400 ± 20 Hz primary power into a 26 volt source and two volt sources, separated in phase by 180°. Testing shall be as specified in 4.5.3.2.

3.3.4 Circuit card assemblies. The circuit card assemblies (CCA) shall incorporate the electrical functions described herein. Testing shall be as specified in 4.5.3.2.

3.3.4.1 Elevation demodulator detector circuit card assembly (A1A5A5). This CCA includes the elevation channel preamplifier and demodulator circuits of the servo system, and the coincidence amplifier. The input signals to the servo preamplifier shall include the sighting system elevation resolver output, the elevation stow adjustment, the pilot range adjustment and the gunner range adjustment. These signals shall be summed and amplified and introduced to the demodulator which shall remove the carrier signal as the first stage in lead compensation signal processing. The scale factor for the preamplifier and demodulator shall be 0.068 vdc per milliradian of error input. One milliradian of error shall be represented by a signal level of 10 millivolts. Therefore a 10 millivolt root mean square (rms), 400 Hz error signal input to the preamplifier Pin D of A1P3 will equal 68 millivolts dc level output from the demodulator Pin M of A1J1. The saturation voltage output shall be 11.0 vdc. This circuit card shall contain a fixed elevation stow bias resistor. The signal inputs described above shall also be introduced to the coincidence amplifier of the noncoincidence detection circuit. These signals shall be amplified and then rectified for presentation to an electronic trigger circuit external to this CCA. The coincidence amplifier scale factor shall be 1.4 vdc per degree of noncoincidence. One degree equals 17.4 milliradians and one milliradian equals 10 millivolts rms, 400 Hz. Therefore, one degree of noncoincidence equals 17.4 x 10 = 174 millivolts. If one degree of noncoincidence occurs, there will be 174 millivolts applied to the input of the coincidence amplifier (Pin D of A1P3). The output of the coincidence amplifier will be at a level of 1.4 vdc (Pin H of A1J1). An error of 5° + 3° - 0° shall be of sufficient magnitude to energize the coincidence relays. This shall be equal to 1.4 x 5 + 3 - 0 = 7 to 11.2 vdc out of the coincidence amplifier.

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3.3.4.2 Azimuth demodulator detector circuit card assembly (A1A5A2). This CCA, which operates on azimuth data, shall be identical to A1A5A5 with the exception that there shall be no range data input in the azimuth channel. The input to the azimuth preamplifier of 10 millivolts (Pin L of A1P3) shall equal 68 millivolts dc level output from the demodulator (Pin L of A1J1). The input to the azimuth coincidence amplifier of 174 millivolts (Pin L of A1P3) shall equal a level of 1.4 vdc (Pin G of A1J1) at the output of the coincidence amplifier.

3.3.4.3 Azimuth equalizer amplifier circuit card assembly (A1A5A3). This CCA contains the azimuth limit circuit, equalizer amplifier and the summing and power amplifier in the azimuth channel. The equalization amplifier provides the appropriate transfer function. The demodulated input signal from (A1A5A2) and an appropriate derivative of this signal shall be provided by the equalization amplifier and introduced to the summing and power amplifier. This signal shall be summed with the amplified tachometer output and amplified to operate the azimuth servo valve. The amplifier provides a differential current over the two halves of the servo valve solenoid. The amplifier shall provide a quiescent current of 5 milliamperes (nominal) through each section of the solenoid when the turret is holding an ordered position and shall be capable of delivering up to a minimum of 10 milliamperes (ma) through one section with zero current in the other section during maximum slew conditions. The equalizer, summing and power amplifier shall have a nominal voltage gain of 0.8 from the equalizer input to a balanced output that is equivalent to the servo valves. (1,000 ohms $\pm$ 100 ohms from each side to ground Pin $\overline{P}$ and $\overline{Q}$ of A1P3.) With a plus 1.0 volt on the equalizer input A1J1-L the output on Pin P will increase 0.4 volt and the output on Pin $\overline{Q}$ will decrease 0.4 volt (nominal values). With an input on the summing and power amplifier A1J1-L, the nominal voltage gain to the output shall be 0.15 total swing. Pin A1P3-$\overline{Q}$ is in phase with A1J1-E and A1P3-$\overline{P}$ is 180° out of phase with A1J1-E. The AZ limit circuit is actuated from the AZ limit switches. Its purpose is to stop or significantly reduce the velocity of the turret prior to its contacting the torque tube hard stop. With a drive signal towards either AZ turret limit (pins A1P30 or $\overline{P}$), the limit switches and the circuitry applies this drive signal in the reverse direction, thus balancing the signal on both halves of the servo valve except for the voltage drop in the circuit. The voltage drop in the circuit shall be 0.1 to 0.4 volts when the circuit is driving hard in either direction (10 ma min in one section of the servo valve and zero in the other section). With anything less than max drive and the limit switch actuated, the limit circuit voltage drop will be reduced and will approach zero for very small drive signals. Reverse input drive signals when either limit switch is actuated should not have any effect on normal operation. Variable resistors are placed in series with the limit switches to compensate for slight differences in the turret servo valves. Increasing these resistors from zero resistance will cause the circuit voltage drop to increase up to 4.0 max with max drive signal input.

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3.3.4.4 Elevation equalizer amplifier circuit card assembly (A1A5A4). This CCA is the elevation channel counterpart of A1A5A3 above, with the exception that the limit circuit is not provided. The output shall drive the elevation servo valve. The same gain and phase relationships shall hold for the elevation equalizer amplifier card as for the azimuth equalizer amplifier, except the output to the servo valves shall be on Pins $\overline{F}$ and $\overline{G}$ of A1P3. With a 1.0 volt P-P on the equalizer input A1J1-M, the output on A1P3-F will increase 0.4 volt and the output on A1P3-$\overline{G}$ will decrease 0.4 volt (nominal values). With an input on the summing and power amplifier A1J1-K the voltage gain to the output shall be 0.418 total swing.

3.3.4.5 Limit-coincidence circuit card assembly (A1A5A1). This CCA contains the azimuth and elevation channel coincidence trigger circuits driven by the rectified output from the azimuth and elevation coincidence amplifiers of A1A5A2 and A1A5A5 above and shall, in the event of non-coincidence, result in operation of the azimuth and elevation coincidence relays. Noncoincidence is defined as $5^{o} + 3^{o} - 0^{o}$ difference between the sight angle and turret angle or error. This angle error is represented by 0.87 to 1.39 volts rms voltage error fed into the demodulation detector circuit cards above. This shall be represented by a 7.0 to 11.2 vdc signal fed into the A1A5A3 and A1A5A4 CCA. This signal shall be of sufficient value to open either of the coincidence relays. This CCA contains the Reticle Flasher circuit which consists essentially of a unijunction transistor oscillator operating nominally at 2 Hz. The output of this flasher shall be capable of actuating the reticle lamp relay whenever a noncoincidence condition occurs. This circuit will complete the ground circuit of the reticle relay for 0.23 $\pm$ 0.06 second and then off for approximately 0.13 second making a total cycle of 0.36 $\pm$ .10 second. This CCA, in connection with the A7 relay bracket assembly, contains an elevation limit circuit that prevents mechanical damage to the turret when the sighting system commands the turret to drive above the turret elevation profile. This circuit develops a 1.0 to 3.0 ma down signal on the two halves of the servo valve between A1P3-$\overline{F}$ and $\overline{G}$, the larger signal being on A1P3-$\overline{F}$. This range of 1.0 to 3.0 ma includes the extremes of the balancing resistor A7R7. The circuit that generates this signal is energized when the turret operates the upper limit switch. Pins A1P3-$\overline{Y}$ and $\overline{X}$ are connected. When the turret comes out of the limit the connection between A1P3-$\overline{Y}$ and $\overline{X}$ is broken and a balance signal is then applied to the two halves of the servo valve, pins P1P3-$\overline{F}$ and $\overline{G}$. This signal is held in balance by adjusting A7R7 at an average of 2.5 $\pm$ .5 ma on both A1P3-$\overline{F}$ and $\overline{G}$. The differential balance range shall be at least $\pm$ 1.0 ma. This may be considered a simulated turret stow position in elevation. The sighting system will continue to have full control of the azimuth position while the turret is in the simulated (elevation) stow position. The turret will stay in the simulated stow position until a

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signal commands the turret, through pins A1P3-$\overline{G}$ and $\overline{F}$, to drive below the simulated stow position. This down drive signal from the sighting station on A1P1-E should be 7.0 + 1 millivolts P.P. in phase with the reference 400 Hz supply. This will result in an unbalance down drive signal between 14R and 15S of the A4 CCA. The voltage on 14R will need to be 20 millivolts larger than 15S in order for the sighting system signal from pin A1P1E to take control of the signal to the turret A1P3-$\overline{G}$ and $\overline{F}$. Removal of the action voltage de-energizes the elevation limit circuit and the turret will return it to normal stow position, zero azimuth and zero elevation. This CCA and A7 relay bracket assembly also contains a second circuit that develops an updrive signal for the turret servo valves (an unbalance between A1P3-$\overline{F}$ and $\overline{G}$ with the larger signal on $\overline{G}$). This updrive signal allows the turret to follow the upper firing profile as the turret returns towards zero azimuth while in the simulated elevation stow position. This updrive signal also exists when the turret is in the simulated elevation stow position and is moving toward the azimuth limits, but it has no beneficial or detrimental effect for this mode of operation. The unbalance signal is generated when the azimuth error signal is increased and shall be present before the error signal at pin A1J1-G exceeds 0.35 volts but shall not be present with 0.20 volts or less error signal on A1J1-G. With larger error signals the unbalance signal will increase and eventually will saturate at 1.0 ma max. This CCA contains an adjustable reference 400 Hz signal source for the system resolver chain. It shall be adjusted to 10.0 $\pm$ .1 volt rms (Pin H of A1P3).

3.3.4.6 Tachometer amplifier circuit card assembly (A1A5A6). This CCA contains the elevation and azimuth tachometer amplifier circuits which receive the tachometer outputs from the azimuth and elevation tachometers in the turret instrument drive assemblies and shall amplify and introduce these outputs to the particluar summing and power amplifier. These amplifiers provide the damping functions for the azimuth and elevation axes. The gain of these amplifiers shall be frequency sensitive. With a sine and a frequency of 0.1 Hz, the voltage gain will be approximately 0.174. With a sine wave and a frequency of 2 Hz, the voltage gain will be 3.5. The inputs for this test shall be Pins X and W of A1P3 with outputs on Pins E and K of A1J1. This CCA contains the azimuth and elevation stow adjustment potentiometers. An adjustment in the output from these potentiometers shall be imposed on the respective amplifiers and shall result in a discrete change in position of the turret in azimuth and elevation. The stow adjustment potentiometers shall both be adjusted for 0 volts rms. This CCA shall contain the fixed range adjustment resistors employed in the elevation resolver output circuit.

3.3.5 Relay functions. The individual relays shall perform the following functions: Testing shall be as specified in 4.5.3.2.

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3.3.5.1 Gunner action relay A1K1. Actuation of this relay shall release the turret from the stow position and permit operation of the turret from the gunners sighting system in either the "SAFE" or "ARMED" position. The A1K1 relay shall be actuated by the application of 26 ± 4 vdc to A1P1-Y. Actuation of A1K1 completes a circuit from A1P1-E to an elevation channel preamplifier circuit, and interrupts the circuit to A1P1-N and A1P3-D to the preamplifier circuit.

3.3.5.2 Compensation relay A1K2. This relay permits the output signals from range potentiometers and air data transducer to be introduced into the servo system for velocity compensation. The A1K2 relay shall be actuated by the application of 26 ± 4 vdc to A1P1-Z. Actuation connects A1P1-G to A1P2-F with a resistor of 10 kilohms between these pins.

3.3.5.3 Arming relay A1K3. The actuation of this relay shall close the circuit from the 26 ± 4 vdc unswitched line to the armed bus in the electronic components assembly, placing the system in the armed state and in readiness for trigger actuation, and the "ARMED" lights shall be on. The A1K3 relay shall be actuated, following actuation of A1K16, by 26 ± 4 vdc applied to any of the following:

A1P4-$\overline{D}$ or $\overline{B}$

A1P5-L or V

Upon actuation of A1K5 by 26 ± 4 vdc applied to A1P5-$\overline{C}$ and 26 ± 4 vdc applied to A1P5-S.

Actuation of A1K3 supplies 26 ± 4 vdc from GG and EE of A1P4 to:

A1P5-W A1P3-U

A1P1-R A1P4-DD, $\overline{Q}$, $\overline{U}$

Upon actuation of A1K7 relay, 26 ± 4 vdc is supplied to A1P1-B from GG and EE of A1P4.

3.3.5.4 Turret power relay A1K4. The contact on this relay shall introduce 26 ± 4 vdc to the turret power bus and to the dc-to-dc converters, thereby permitting control of the turret with the system in the "SAFE" mode. The A1K4 relay shall be actuated following actuation of A1K16 and one of the following conditions:

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Upon actuation of	With 26 ± 4 vdc applied	Interrupted by actuation of
-	A1P4-$\overline{Z}$	A1K5
A1K8	A1P1-A	
A1K15	A1P4-$\overline{T}$	
-	A1P1-$\overline{A}$	A1K5 or A1K15
-	A1P5-$\overline{D}$	
A1K5	A1P5, Pin X, C and Y	
-	A1P4-$\overline{G}$	
A1K11 and A1K12	A1P3-S	Either A1K11 or A1K12 de-energized

The A1K4 relay shall remain actuated while 26 ± 4 vdc is applied to either of the following:

Terminal	Interrupted by actuation of
A1P3-$\overline{R}$	-
A1P5-$\overline{J}$	A1K3
A1P4-AA	A1K3

3.3.5.5 Pilot override relay A1K5. This relay shall close the circuit from the 26 ± 4 vdc unswitched line to the gunner control bus and the armed bus and shall simultaneously open-circuit the pilot controls and sighting system triggers thereby permitting the gunner to override the pilot. This relay shall open the circuit to the "PILOT IN CONTROL" indicator and simultaneously energize the "GUNNER IN CONTROL" indicator. The A1K5 relay shall be actuated by 26 ± 4 vdc applied to A1P5-$\overline{C}$.

Actuation of A1K5 connects:

A1P5-S to A1P5-V and L

A1P5-S to A1P4-$\overline{B}$

26 ± 4 vdc to A1P4-$\overline{G}$, A1P5-$\overline{D}$

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3.3.5.6 Gunner control relay A1K7. The contacts of this relay shall close the circuit from the armed bus to the trigger switches and open the pilot controls, thereby transferring armament control from the pilot to the gunner.

The A1K7 relay shall be actuated by any of the following conditions:

Upon actuation of	With 26 ± 4 vdc applied
-	A1P5-$\overline{D}$
-	A1P4-$\overline{G}$
A1K5	A1P5 Pins, C, S and Y

When actuated, A1K7 connects:

A1P1-L to A1P4-R	A1P1-F to A1P4-D
A1P1-L to A1P5-$\overline{H}$	A1P1-F to A1P5-N

3.3.5.7 Reticle relay A1K8. This relay shall remain energized when the turret is properly aligned with the sight thereby providing continuous operation of the reticle lamp. When the noncoincidence occurs, this relay shall be operated by the flasher circuit thereby resulting in flashing of the reticle lamp. The A1K8 relay shall be in a noncoincidence condition when 7.0 to 11.2 vdc is applied to either A1J1-F or A1J1-G. Coincidence state shall apply when 6.4 vdc or less is applied to A1J1-H or A1J1-G. Actuation of A1K8 supplies 26 ± 4 vdc to A1P1-$\overline{E}$ from Pin GG or EE from A1P4.

3.3.5.8 Burst limit relay, six seconds A1K9. This relay shall have a fixed time delay upon receipt of the operate signal of six seconds to limit the burst time of the 7.62mm weapon.

The A1K9 relay shall be actuated by either of the following:

Upon actuation of	With 26 ± 4 vdc applied	Interrupted by actuation of
-	A1P4-G	A1K9 (relay may cycle)
A1K12 and A1K11	A1P3-$\overline{K}$	A1K14 or de-energize A1K11 or A1K12

3.3.5.9 40MM burst limit relay, ten second A1K10. This relay shall be actuated by either of the following:

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Upon actuation of	With 26 ± 4 vdc applied	Interrupted by actuation of
-	A1P4-$\overline{I}$	A1K10 (relay may cycle)
A1K14, A1K12, A1K11	A1P3-$\overline{K}$	De-energize A1K14, A1K12, or A1K11

3.3.5.10 Azimuth coincidence A1K11. This relay shall be de-energized when the signal resulting from azimuth noncoincidence attains the required level. The relay action shall switch the reticle relay (A1K8) coil into the flasher circuit, and shall interrupt the firing circuitry. A1K11 relay is normally energized.

3.3.5.11 Elevation coincidence A1K12. This relay shall de-energize during elevation noncoincidence and shall produce the same result as noted for A1K11 above. A1K12 is normally energized. A1K12 shall also be energized from a 28 volt signal applied to Pin $\overline{D}$ of A1P1 which simulates a condition where the gunner's sight trigger switch is pulled before the action switch to prevent firing before the out of coincidence error can be acted upon.

3.3.5.12 7.62mm, 40mm firing A1K13 and 40mm firing A1K14. These relays are operated in parallel to provide sufficient contacts to perform the required functions. These relays shall be energized when firing a 40mm weapon or when two 40mm weapons are installed. A1K14 shall assist the trigger logic circuitry. Relay A1K13 shall simultaneously replace the 7.62mm range compensation resistors with the 40mm compensation resistors in the range adjustment circuits when a 40mm weapon is being fired.

The A1K13 and A1K14 relays shall be actuated by any of the following conditions:

Upon actuation of	With 26 ± 4 vdc applied	Interrupted by actuation of
-	A1P3-T	-
A6K3	A1P1-S	A6K1 or A6K3 de-energized

The action of A1K13 supplies 26 vac, 400 Hz to A1P2-E.

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3.3.5.13 Pilot control A1K15. Actuation of this relay shall permit operation of the armament by the pilot (see 3.3.5.6).

The A1K15 relay shall be actuated by any of the following:

Upon actuation of	With 26 ± 4 vdc applied	Interrupted by actuation of
-	A1P4-$\overline{Z}$	A1K5
A1K8	A1P1-A	A1K8 de-energized

The actuation of A1K7 relay or the A1K15 relay shall apply 400 Hz to the power transformer A1T1.

3.3.5.14 Fail safe A1K16. This relay shall be de-energized upon loss of the 115 vac, 400 Hz at A1P4-X. The relay drop-out shall de-energize the arming A1K3 and turret power A1K4 relays thereby disarming the system and providing a controlled shut-down of the turret action.

3.3.5.15 7.62mm trigger interlock relay A6K1. The contacts of this relay shall open-circuit the 40mm trigger circuit for the purpose of establishing trigger priority.

The A6K1 relay shall be actuated by the following:

Upon actuation of	With 26 ± 4 vdc applied	Interrupted by actuation of
-	A1P-$\overline{D}$	A6K3

3.3.5.16 Gunner action relay A6K2. This relay shall operate in parallel with A1K1 (see 3.3.5.1) for the purpose of providing additional contacts. The contacts shall introduce the pilot range compensation signal into the elevation preamplifier input of CCA A1A5A5.

The A6K2 relay shall be actuated by 26 ± 4 vdc applied to either:

A1P1-Y or A1P5-A

3.3.5.17 40mm trigger interlock relay A6K3. This relay shall operate in a manner similar to A6K1, for the purpose of establishing trigger priority. The relay contacts shall open-circuit the 7.62mm trigger circuit.

The A6K3 relay shall be energized by:

Upon actuation of	With 26 ± 4 vdc applied	Interrupted by actuation of
-	A1P1-S	A6K1

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3.3.5.18 Elevation limit circuit actuation Relay A7K1 and A7K2. These relay coils are connected in parallel for the purpose of providing additional contacts. These relays energize the elevation circuit described in paragraph 3.3.4.5. Basically, they shift the control of the elevation servo valve from the servo loop control to a fixed dc source. When energized they cause the turret to move down, and just out of the elevation limit. When A7K1 and A7K2 are de-energized, and the A5A4 CCA is removed, there shall be continuity between the following pins: A5A4-15S and A1P3-$\overline{G}$, A5A4-14R and A1P3-$\overline{F}$. Also, there shall be 1000 $\pm$ 100 ohms between A5A1-7H and A5A1-R. With 26 $\pm$ 4 vdc applied to A1P3-$\overline{Y}$ and ground, A7K1 and A7K2 shall be energized. With 26 $\pm$ 4 vdc applied to A1P5-A and ground, the voltage can be removed from A1P3-$\overline{Y}$, relay A7K1 and A7K2 shall remain energized. With A7K1 and A7K2 energized there shall be continuity between the pins A5A4-15S and A5A1-A, A5A4-14R and A5A1-1, A5A4-13P and A5A4-2B, A1P3-DD and A1P3-EE. With A7K1 and A7K2 energized from A1P5-A, 26 $\pm$ 4 vdc can be measured at A1P3-$\overline{F}$ and A1P3-$\overline{G}$ less the circuit voltage drop.

3.3.5.19 Elevation limit circuit de-activating relay A1K3. When a down drive signal is transmitted from the sighting system (see 3.3.4.5), the A1K3 relay is energized. When the A1K3 is energized, the A7K1 and A7K2 are de-energized. The sighting-system signal on A1P1-E will again have control of the turret which is the signal on A1P3-$\overline{G}$ and A1P3-$\overline{F}$.

3.3.6 Voltage out. Unswitched 26 $\pm$ 4 vdc is supplied to terminals A1P5-C, A1P5-$\overline{X}$, A1P5-$\overline{Y}$ and A1P4-J when 26 $\pm$ 4 vdc is applied to A1P4-GG and A1P4-EE.

3.4 Environmental conditions. The environmental conditions shall be as specified in the contract (see 6.2). Testing shall be as specified in 4.4 and 4.6.3.

3.5 Reliability. When specified in the contract (see 6.2), the ECA shall have a minimum acceptable Mean-Mission-Between-Failure (MMBF) of not less than 75 missions with a desired and specified MMBF of 112 missions. A mission is defined as those functions of the electronic components assembly necessary to simulate the expenditure of 4,000 rounds of 7.62mm and 265 rounds of 40mm ammunition during a two-hour flight. A failure will be defined as a nonoccurrence of the appropriate event when power is applied and the required control switches are actuated. Testing shall be as specified in 4.4 and 4.6.4.

3.5.1 Reliability criteria. The above reliability requirements are based on a minimum acceptable Mean-Time-Between-Failure (MTBF) of 150 hours.

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3.6 Electromagnetic interference (EMI). The electromagetic interference (EMI) shall be as specified in the contract (see 6.2). Testing shall be as specified in 4.4 and 4.6.5.

3.7 Interchangeability. The six CCAs shall be interchangeable with CCAs of the same part number. Testing shall be as specified in 4.5.3.3.

3.8 Marking. Each ECA, CCA and parts for which markings are specified shall be clearly marked in accordance with the applicable drawings and MIL-W-13855. Each ECA and CCA shall be identified by a serial number assigned by the procuring activity (see 6.2).

3.9 Workmanship. Workmanship shall be in accordance with MIL-W-13855.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality assurance terms and definitions. Quality assurance terms and definitions used herein are in accordance with MIL-STD-109 and MIL-STD-721.

4.3 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.4 First article inspection. The first article (initial production unit(s)) shall be submitted for inspection in accordance with the contract (see 6.2). The first article shall be representative of the production processes to be used during quantity production. The first article shall be subjected to the quality conformance inspection specified herein and, when specified in the contract (see 6.2), the environmental conditions test (see 3.4 and 4.6.3), reliability test (see 3.5 and 4.6.4), electromagnetic interference test (see 3.6 and 4.6.5) and such other inspections as is necessary to determine compliance with the requirements of the contract.

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4.5 Quality conformance inspection.

4.5.1 Inspection lots. The formation, size and presentation of inspection lots shall be in accordance with MIL-STD-105. Inspection lots shall be as large as practicable, in consideration of quality history, manufacturing conditions and the contractor's delivery schedule and within limitations of MIL-W-13855.

4.5.2.1 Component parts and concurrent repair parts. Examination of parts shall be performed in accordance with the criteria specified in the contract (see 6.2). The contractor's examination of parts shall be accomplished prior to their assembly into the end items or submission for acceptance as repair parts.

4.5.2.2 Electronic components assembly. Each ECA shall be visually and manually examined to determine conformance with the material, construction and design requirements of 3.2 and 3.2.1. Each step in the examination shall include a visual examination for proper cleaning and presence of the specified protective coating and to determine the general quality, completeness of manufacture, assembly and workmanship. The examination provisions shall be applied at the earliest practical point in manufacture at which it is feasible to inspect for acceptance without risk of change in the characteristic by subsequent operations. Reinspections of these characteristics on the completed product is not required provided assurance exists that the characteristic has not been changed, degraded or damaged by subsequent manufacturing, assembly or handling, and that adequate inspection records are maintained. ECAs failing to meet the requirements shall be rejected.

4.5.2.2.1 Visually examine all markings required by the final assembly drawing to assure they are clear and legible.

4.5.2.2.2 Visually examine the identification plate for compliance with the applicable drawing, for secure assembly (see 3.8), serialization, and to assure that all pertinent data has been applied.

4.5.3 Testing.

4.5.3.1 Failure data. Unless otherwise specified herein, all tests shall be conducted on a complete ECA. If test requirements specified herein are not met, acceptance of ECAs shall be deferred and the contractor shall accomplish, as applicable, the following actions:

a. Conduct a failure analysis study performing a dimensional, physical and visual examination of the components which are suspected to be the cause of failure or malfunction.

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b. Evaluate and correct the applicable production processes and procedures to prevent recurrence of the same defect(s) in future production.
c. Examine ECAs, partially assembled ECAs and components (including components and subassemblies at in-process or final assembly) to insure that material containing the same defect is purged from the inventory and not presented to the Government for acceptance.
d. Submit the results of the failure analysis and the corrective actions taken to the Government for review and approval prior to submitting a reconditioned lot or reconditioned ECAs for retest.

4.5.3.2 Functional testing. Each electronic component assembly shall be tested in accordance with the test method specified in 4.6.1 to determine compliance with functional requirements of 3.3.1 through 3.3.6. Electronic components assemblies failing to meet the functional requirements shall be rejected.

4.5.3.3 Interchangeability testing. The supplier shall test a sample of three electronic components assemblies from each month's production for interchangeability using the test method specified in 4.6.2. Electronic components assemblies taken for interchangeability shall have been found satisfactory in all other examinations and tests, including function test. The electronic components assemblies shall be function tested (see 4.6.1) after interchange of parts. Failure of the interchangeability test shall be cause for rejection of the entire lot. Prior to submission as a reconditioned lot, the contractor's correction shall be effected on all electronic components assemblies of the lot and on all electronic components assemblies and component parts not yet accepted. Sample size and test method for reconditioned lots shall be the same as for the original lot.

4.5.3.4 Component parts and concurrent repair parts testing. Raw material testing, part testing and certification shall be performed in accordance with the criteria specified in the contract (see 6.2). This will include chemical analysis and physical tests of materials, tests of protective finish, heat treatment, bonding and function of parts as applicable. The contractor shall accomplish these tests prior to assembly of parts into the end item.

4.5.4 Packaging examination and testing. Unless otherwise specified (see 6.2), the packaging examination and testing shall be in accordance with MIL-P-14232.

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4.5.5 Inspection equipment. Unless otherwise specified (see 6.2), responsibilities for acquisition, maintenance and disposition of inspection and test equipment prescribed on Quality Engineering Data List QE-G-DL 11691622 and for all other inspection equipment required to perform inspection required by applicable specifications, shall be in accordance with MIL-I-45607.

4.6 Test methods.

4.6.1 Functional test. The functional test shall be performed in accordance with the test procedure specified on QE-G-DL 11691622 (see Drawing 11836897).

4.6.1.1 Degradation test. Degradation test may be performed in conjunction with functional test (see 4.6.1). Degradation test shall be conducted with the dc voltage level reduced to 22.0 + 0.5 vdc. Degradation shall be within the requirements specified in 3.3.1.

4.6.2 Interchangeability test. The electronic components assemblies shall be tested for interchange of circuit card assemblies in the following manner, to determine compliance with 3.7. The three units selected for interchangeability test shall be designated as Numbers 1, 2, and 3. From unit one, circuit card assemblies (CCAs) 11688129 and 11688131 shall be interchanged with the same part numbers from unit two. From unit one, CCAs 11688121 and 11688128 shall be interchanged with the same part numbers from unit three. After interchange of CCAs is accomplished, each reassembled ECA shall be functionally tested in accordance with the test method in 4.6.1.

4.6.3 Environmental conditions test. When specified in the contract (see 6.2), electronic components assemblies shall be subjected to and environmental conditions test.

4.6.4 Reliability test. When specified in the contract (see 6.2), the reliability test shall be performed by repeating the test cycle defined in 4.6.4.1. For each test cycle, the failure versus test cycles data shall be cumulatively plotted. Testing shall be terminated whenever a decision point as defined in Figure 1 has been reached.

4.6.4.1 Reliability test cycle. One test cycle (representing the functions necessary to perform one mission) shall be completed by performing the procedures contained in the following paragraphs from Test Procedure 11836897 (see QE-G-DL 11691622) used to demonstrate the requirements of 3.5:

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Paragraph No.	Repetitions	Characteristics Tested
4.1, 4.2, 4.3	-	Preparation for Test
5.1, 5.2	-	Test Conditions
5.3	5	Arming Relay A1K3 Turret Power Relay A1K4 Pilot Override Relay A1K5
5.3	5	Pilot Control Relay A1K5 Fail Safe Relay A1K16 7.62 Trigger Interlock Relay A6K1 40mm Trigger Interlock Relay A6K3
5.3	5	Turret Power Relay A1K4 Pilot Control Relay A1K15 Fail Safe Relay A1K16
5.9	1	Reticle Relay A1K8
5.10	5	6 Second Burst Limit Relay A1K9 7.62-40mm Firing and 40mm Firing Relay A1K14
5.11	5	10 Second Burst Limit Relay A1K10
5.13	12	Azimuth Coincidence A1K11 Elevation Coincidence A1K12
5.12	3	Gunner Action Relay A1K1 Gunner Action Relay A6K2
5.3	5	Gunner Control Relay A1K7
5.17.1	5	7.62-40mm Firing and 40mm Firing Relay A1K14
5.3	17	Compensation Relay A1K3

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4.6.4.2 The contractor shall provide replacement parts as required to complete the testing.

4.6.4.3 At the option of the contractor, the tests may be interrupted during, or at the end of a cycle and be restarted later at the point at which it was interrupted.

4.6.4.4 Complete, accurate records shall be kept for each reliability test, showing each failure and part replacement, including an indication of when each failure occurred. In addition, complete, accurate records shall be kept for all time losses attributable to the electronic components assemblies, including the time for each failure.

4.6.5 Electromagnetic interference test. When specified in the contract (see 6.2), electronic components assemblies shall be subjected to an electromagnetic interference test.

5. PACKAGING

5.1 Pilot pack. A pilot pack shall consist of a complete electronic components assembly preserved in accordance with Packaging Data Sheet P11691622 for the level of protection specified in the contract (see 6.2), packed Level B and forwarded as specified in 3.1.

5.2 Levels A and B. Preservation, packing and marking shall be in accordance with Packaging Data Sheet P11691622 for the level of protection specified in the contract (see 6.2).

5.3 Commercial packaging. Preservation, packing and marking shall be in accordance with Fed. Std. No. 356.

5.4 Repair parts. Repair parts shall be preserved, packed and marked in accordance with the Packaging Data Sheet or other packaging requirements, as specified in the contract (see 6.2).

6. NOTES

6.1 Intended use. The electronic component assembly is intended for use on, but not limited to, the M28A1E1 Helicopter Armament Subsystem.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. List of drawings, specifications and publications pertinent to the weapon system showing applicable revision dates.
c. Shipping instructions for first article and pilot pack (see 3.1, 4.4 and 5.1).

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d. Environmental conditions, reliability and electromagnetic interference requirements and tests, when required (see 3.4, 3.5, 3.6, 4.4, 4.6.3, 4.6.4 and 4.6.5).
e. Block of serial numbers for ECAs (see 3.8).
f. Block of serial numbers for CCAs (see 3.8).
g. Examination criteria for component parts and concurrent repair parts (see 4.5.2.1).
h. Testing criteria for component parts and concurrent repair parts (see 4.5.3.4).
i. Packaging examination and testing, if different (see 4.5.4).
j. Responsibilities for acquisition, maintenance and disposition of inspection equipment, if other than as specified (see 4.5.5).
k. Selection of applicable level of protection (A or B), or commercial packaging (see 5.1, 5.2 and 5.3).
l. Packaging instructions for repair parts (see 5.4).

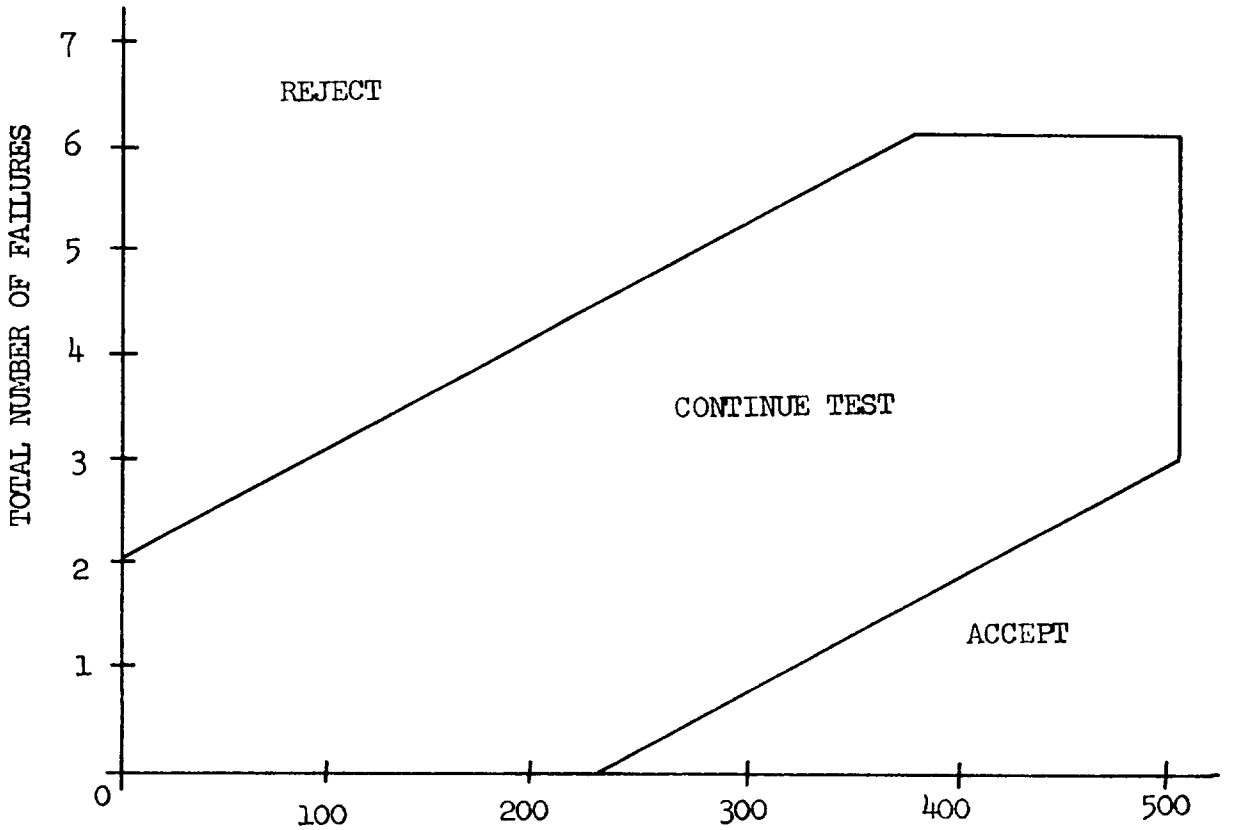
Custodian:
Army - WC

Preparing activity:
Army - WC

Project Numbers:
(1090-A026)

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Decision Risks: 30%
Discrimination Ratio: 1.5: 1
MIL-STD-781 Test Plan VII



NO. OF FAILURES	REJECT (EQUAL OR LESS)	ACCEPT (EQUAL OR MORE)
0	N/A	236.3
1	N/A	327.4
2	N/A	418.5
3	91.1	509.6
4	182.3	509.6
5	273.4	509.6
6	509.6	N/A

FIGURE 1

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No 22-R255

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DOCUMENT IDENTIFIER AND TITLE

MIL-E-45962(WC); Electronic Components Assembly (11691622)

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3 IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4 REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO
	DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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